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Neguse Solomon Mekonnen

Nationality: Ethiopian **Date of birth:** 31/05/2000 **Gender:** Male

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ABOUT ME

Civil and software engineer and structural reliability researcher specializing in probabilistic assessment and uncertainty quantification of concrete structures. Erasmus Mundus MSc graduate (UPC Barcelona, University of Padua) and BSc Computer Science candidate, seeking a PhD in probabilistic computational modelling and structural assessment of existing structures.

EDUCATION AND TRAINING

MSc Erasmus Mundus Joint Masters — NORISK: Risk Assessment and Management of Civil Infrastructures

UPC Barcelona + Università degli Studi di Padova

[Oct 2025 – Sep 2026]

City: Barcelona / Padua | **Country:** Spain / Italy

- **Thesis:** “Bayesian Updating of Bridge Structural Reliability Using Monitoring Data, Stop Criteria, and Proof Load Testing: A Case Study of the Zijlweg Viaduct”
- **Supervisors:** Dr. Eva Lantsoght (TU Delft/USFQ) and Dr. Rolando Chacón (UPC Barcelona)
- **Methods:** Bayesian MCMC (Metropolis-Hastings, 10^8 iterations in C++), Latin Hypercube Sampling, CSCT/CSDT stop criteria, cross-section analysis with Thorenfeldt concrete model
- **Contribution:** First field application of the De Vries (2025) Bayesian reliability updating framework to real proof load test data from a deteriorated ASR-affected structure

BSc in Computer Science (in progress)

University of the People

[Sep 2024 – Jan 2027 (expected)]

City: Wilmington | **Country:** United States | **Study mode:** Online

BSc in Civil Engineering — First Class Honours

Wollo University

[Oct 2018 – Sep 2023]

City: Kombolcha | **Country:** Ethiopia | **Final grade:** 3.41 / 4.0

- **Thesis:** “Structural Design and Performance Optimization of a Mixed-Use Building Integrating Sustainable Geopolymer Concrete via Building Information Modeling (BIM)”
- Graduated with Distinction; commenced university teaching and structural laboratory research from 3rd year onward — two years before degree completion

WORK EXPERIENCE

Lecturer — Structural Analysis & Computational Engineering

Wollo University, Department of Civil Engineering – Kombolcha, Ethiopia

[Sep 2023 – Present]

- Deliver undergraduate lectures in structural analysis, reinforced concrete design, and computational engineering tools (ETABS, SAP2000, AutoCAD) — courses that directly underpin probabilistic structural assessment
- Design course materials and assessment rubrics; maintain 90%+ student success rate; mentor undergraduate research students from problem formulation through quantitative analysis

Technical Advisor

Ethiopian Youth Entrepreneurs Association (EYEA) – Addis Ababa, Ethiopia

[Jan 2024 – Present]

- Deliver technical mentorship on structural design, load analysis, and documentation to emerging construction entrepreneurs; provide structural drawings and design reviews

Academic Tutor & Laboratory Assistant

Wollo University, Department of Civil Engineering – Kombolcha, Ethiopia

[Oct 2021 – Sep 2023]

- Supported professors in structural mechanics, soil mechanics, and structural analysis for 200+ students per year (90%+ course success rate); operated structural and geotechnical laboratory equipment and prepared laboratory manuals and tutorial materials formally adopted into course curricula

RESEARCH EXPERIENCE

Graduate Researcher — Structural Reliability

UPC Barcelona / TU Delft (Dr. Eva Lantsoght)

[Mar 2026 – Sep 2026]

- Applied the De Vries (2025) Bayesian MCMC framework to field monitoring data from the 2015 proof load test of Viaduct Zijlweg, Netherlands — first such field application to a real ASR-deteriorated bridge rated analytically insufficient
- Implemented cross-section analysis in C++ using Latin Hypercube Sampling (100 samples), producing moment-strain curves and yield moment statistics: $M_{y,mean}$ = 3,248.87 kNm, CoV = 0.137, with Thorenfeldt concrete and QR22/QR24 plain-bar steel models
- Quantified monitoring data advantage as +1.85 reliability index units over the uninstrumented lower-bound approach at the same proof load level — first numerical quantification for a real deteriorated structure
- Implemented CSCT and CSDT stop criteria alongside Bayesian updating, demonstrating complementary consistency between the probabilistic and operational safety frameworks
- Preprocessed proof load test data in MATLAB; performed convergence diagnostics and generated all research figures in Python (numpy, scipy, matplotlib, pandas)

Research Assistant — Structural and Materials Testing Laboratory

Wollo University, Department of Civil Engineering – Kombolcha, Ethiopia

[Jan 2023 – Present]

- Conduct concrete, steel and aggregate characterisation tests and controlled structural member load tests, collecting strength and load-deformation datasets used to calibrate probabilistic capacity and analytical resistance models
- Maintained continuously, including remotely and during return visits, while completing international studies

PUBLICATIONS AND RESEARCH OUTPUT

Conference Paper — Accepted and Presented, 22nd International Probabilistic Workshop (IPW 2026); Proceedings Forthcoming

Mekonnen, N.S., Nogueira de Resende, P., & Osuagwu, M.C. (2026). “Multi-Hazard Risk Framework for Building Stock: A GIS-Based Prioritization Methodology against Seismic, Flood, Wind, and Fire Hazards.” In Proceedings of the 22nd International Probabilistic Workshop (IPW 2026). Springer, Lecture Notes in Civil Engineering (forthcoming).

Manuscript in Preparation — Target: Engineering Structures (Elsevier, Q1)

Mekonnen, N.S., Lantsoght, E.O.L., Chacón, R. (2026). “Linking crack-width stop criteria to structural reliability in proof load testing.”

MSc Dissertation

Mekonnen, N.S. (2026). Bayesian Updating of Bridge Structural Reliability Using Monitoring Data, Stop Criteria, and Proof Load Testing: A Case Study of the Zijlweg Viaduct. Universitat Politècnica de Catalunya (UPC), Barcelona.

SELECTED RESEARCH PROJECTS

Forensic Assessment of Urban Flexible Pavement Failure and Condition-Based Maintenance Optimization: A Case Study of Kombolcha City

Wollo University

[May 2023 – Aug 2023]

- Diagnosed pavement failure mechanisms by linking distress patterns to drainage deficiencies, material properties, and traffic loading.
- Developed evidence-based, condition-based maintenance recommendations to improve pavement serviceability and asset longevity.

SKILLS

Research Methods: Bayesian inference, MCMC (Metropolis-Hastings), structural reliability analysis, uncertainty quantification, Latin Hypercube Sampling, limit-state modelling, probabilistic simulation, CSCT/CSDT stop criteria

Computational: C++ (high-performance MCMC, structural analysis), Python (numpy, scipy, matplotlib, pandas), MATLAB (data processing, signal analysis, convergence diagnostics)

Software Engineering: Computer science foundations (BSc in progress); algorithm design and numerical implementation; high-performance C++ for probabilistic simulation; Python scientific computing and data analysis; data-processing pipelines for test and monitoring data

Structural Analysis: Cross-section analysis (Thorenfeldt concrete model, plain bar steel), moment-strain curves, yield moment statistics, proof load test data interpretation, reinforced concrete design

Engineering Software: ETABS, SAP2000, SAFE, STAAD.Pro, AutoCAD, Civil 3D, Revit, Tekla

GIS and Risk: ArcGIS, QGIS — multi-hazard risk mapping, building stock assessment (seismic, flood, wind, fire), normalized risk index development

Laboratory: Concrete compressive/tensile testing, steel mechanical testing, aggregate characterisation, structural load testing, load-deformation data collection

LANGUAGE SKILLS

Mother tongue(s): Amharic

English: Professional working proficiency — scientific writing, research, technical documentation, international academic presentation

PROFESSIONAL MEMBERSHIPS AND LEADERSHIP

- American Society of Civil Engineers (ASCE) — Member
- Ethiopian Association of Civil Engineers (EACE)
- ECESA Wollo Chapter President (Nov 2019 – Sep 2023) — administered student association of 1000+ members, directed policy, managed executive team
- Volunteer Teacher, Addis Ababa City Administration Education Bureau — mentorship programme for underprivileged students (Jun – Sep 2019)

REFERENCES

Available upon request.